

A Study on the Impact of Dynamic Pricing Strategies Based on Deep Reinforcement Learning on Customer Loyalty

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Abstract

Against the backdrop of the rapid development of big data technology, the phenomenon of price discrimination, colloquially known as “big data-enabled price gouging,” has sparked widespread social concern and a crisis of consumer trust. To investigate this issue and seek solutions, this study combines traditional empirical analysis with cutting-edge machine learning simulation. First, through questionnaire surveys and regression analysis, and based on equity theory and price sensitivity theory, we verify the direct negative impact of big data-driven price discrimination on customer loyalty, revealing the mediating role of perceived unfairness and the moderating role of price sensitivity. Second, based on these empirical results, we construct a high-fidelity market simulation environment and employ the Deep Q-Network (DQN) algorithm from Deep Reinforcement Learning (DRL) to train a dynamic pricing agent. The agent’s objective is to maximize long-term profit while maintaining customer loyalty. The simulation results show that the trained reinforcement learning strategy significantly outperforms both myopic greedy pricing and fixed-price strategies in terms of total revenue and customer retention. The final policy learned by the agent confirms that to maintain long-term customer relationships, it is necessary to offer discounts to price-sensitive, low-loyalty users while adopting moderate pricing for high-loyalty users, thereby effectively validating our initial theoretical hypotheses. This study not only reveals the internal mechanism through which price discrimination affects customer loyalty but also provides a scientific basis and practical path for enterprises to optimize pricing and achieve sustainable development using advanced algorithms.

CCS Concepts

• **Applied computing** → Electronic commerce; Online shopping.

Keywords

Deep Reinforcement Learning, Dynamic Pricing, Customer Loyalty, Perceived Unfairness, Price Sensitivity, Deep Q-Network (DQN)

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1 Introduction and Theoretical Basis

The popularization of big data technology is quietly changing the current business ecosystem, pushing enterprises to shift from experience-based judgment to data-driven decision-making models. Algorithmic technology can capture consumer behavior in real-time and convert it into specific quantitative information, which not only improves operational efficiency but also, to some extent, changes the market supply-demand relationship. However, some e-commerce platforms use their data advantages to implement differentiated pricing based on users’ consumption capacity and purchasing habits, a practice known as “big data-enabled price gouging.”^[9,10] This short-sighted algorithmic pricing, while pursuing short-term profit maximization, severely damages consumer rights, undermines fair market competition, and ultimately erodes the company’s most valuable asset—customer loyalty^[2–5].

Traditional empirical research has mostly used surveys or regression analysis to reveal the negative “consequences” of price discrimination but has failed to provide forward-looking “solutions.” How can a smarter, more far-sighted pricing mechanism be designed to find the optimal balance between corporate profits and customer relationship maintenance? To answer this question, this study introduces the Reinforcement Learning (RL) method from the field of artificial intelligence^[7], aiming to provide a new, actionable theoretical and practical framework for resolving the “big data-enabled price gouging” dilemma by combining causal inference from traditional social sciences with the optimization decisions of machine learning^[1–3].

The theoretical framework of this study is primarily based on Equity Theory and Price Sensitivity Theory. Equity Theory posits that consumers compare the ratio of their own inputs (e.g., price paid, time spent) to outputs (e.g., goods or services received) with the ratio of a reference object (e.g., other consumers or their own past experiences). When consumers perceive their ratio to be lower than that of the reference object, a sense of unfairness arises, which may lead them to take actions to restore psychological balance, such as reducing their inputs (e.g., lowering purchase frequency, ceasing purchases)^[6]. This directly forms the theoretical basis for

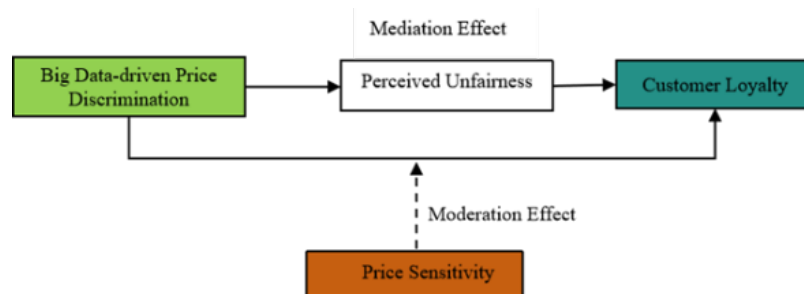


Figure 1: Theoretical model

our H1 (price discrimination has a negative impact on loyalty) and H2 (perceived unfairness acts as a mediator).

Based on equity theory and price sensitivity theory^[6], the theoretical model (as shown in Figure 1) and core hypotheses of this study are as follows:

- H1: Big data-driven price discrimination has a significant negative impact on customer loyalty.
- H2: Perceived unfairness plays a mediating role in the relationship between big data-driven price discrimination and customer loyalty.
- H3: Price sensitivity plays a moderating role in the impact of price discrimination on customer loyalty.

Through the review of existing literature, it can be found that the intensity of big data-driven price discrimination as the independent variable, and its price difference and transparency directly affect customer loyalty. Among them, explicit price difference triggers consumers' perceived unfairness through direct price comparison, while implicit discrimination exacerbates consumers' trust crisis through algorithmic opacity. This dual impact mechanism is reflected in the dimensions of consumer repurchase and brand preference. Procedural fairness and outcome fairness, as the core components of perceived fairness, reflect consumers' judgment of the fairness of the algorithmic decision-making process and outcome, respectively. Price sensitivity, as a moderating variable, reveals that the rejection of algorithmic pricing strategies by high price-sensitive groups is significantly higher than that of low price-sensitive groups. Figure 1 demonstrates this theoretical model.

2 Research Method

2.1 Data Source and Preprocessing

2.1.1 Relationship between big data-driven price discrimination and customer loyalty. The data required for the machine learning modeling in this study was sourced from the first phase of empirical research. We conducted a survey of e-commerce and online travel platform users aged 18-55 using stratified sampling through an online questionnaire platform. The questionnaire consisted of 22 structured questions, using a five-point Likert scale to measure four core variables: the intensity of big data-driven price discrimination, perceived unfairness, customer loyalty, and price sensitivity.

Before using the survey data to construct the simulation environment, we performed data preprocessing. This mainly included:

1) Data Cleaning: Removing invalid or incomplete responses to ensure data quality. 2) Data Normalization: Normalizing the Likert scale scores into continuous values between 0 and 1 to serve as core attributes for the simulated customers, such as initial loyalty and price sensitivity.

2.1.2 Variable measurement and reliability analysis.

Variable Measurement. A five-level Likert scale (1=completely disagree, 5=completely agree) was used, and the scale design was referred to the consumer intention to continuously use scale and adjusted in combination with the pre-study.

- ① Intensity of big data-driven price discrimination, containing two dimensions of explicit and implicit, totaling 4 question items.
- ② Perceived unfairness, integrating procedural fairness and outcome fairness, with 4 question items.
- ③ Customer loyalty, including behavioral loyalty and attitudinal loyalty, totaling 6 items.
- ④ Price sensitivity, including sensitivity to price and the importance of price in the purchase decision, totaling 4 question items.

The final scale is shown in Table 1.

Reliability Analysis. The study conducted Cronbach's α coefficient test on the scales of three variables which are big data-driven price discrimination intensity, perceived unfairness and customer loyalty. The specific results are shown in Table 2. The Cronbach's α coefficient of each scale is significantly higher than 0.7, indicating that the scales have good internal consistency. The reliability meets the research requirements and is suitable for subsequent in-depth analysis.

Validity analysis. Validity refers to the extent to which the measurement instrument can accurately reflect the connotation of the research variables, and is an important basis for assessing whether the study effectively characterizes the target concept. Table 3 presents the KMO measure values of the questionnaire with the results of Bartlett's sphericity test. The KMO value of the questionnaire in this study is higher than 0.7 and the significance level of the Bartlett's spherical test reaches a statistically significant level ($p < 0.05$), which indicates that the questionnaire data has a good structural validity and is suitable for further analysis.

Table 1: Scale item setting

Variable	Item
Big data-driven price discrimination	1. The platform often shows me higher prices. 2. I feel that the platform charges higher prices for old users. 3. I notice that the platform sets different prices for different levels of users. 4. I notice that the platform hides the low-priced goods entrance for long-term members.
Perceived unfairness	1. The online reviews of the store's products are fair. 2. The online reviews of the store's products are objective and unbiased. 3. Online reviews of the store's products are complete. 4. Online reviews of the store's products are correct.
Customer loyalty	1. I believe the platform does not use data to unfairly price users. 2. I often buy goods or services from the platform repeatedly. 3. I am willing to try new products or services from the platform. 4. I have a strong emotional attachment to the platform. 5. I am willing to recommend the platform to others. 6. I have a high sense of identity with the platform's brand.
Price sensitivity	1. I usually change my purchase decision because of the price difference. 2. I spend a lot of time comparing the prices of different platforms. 3. I am very sensitive to changes in product prices. 4. Price is the primary factor for me to choose the platform.

Table 2: Reliability analysis of each scale

Variables	Number of items	Cronbach's coefficient
Intensity of big data-driven price discrimination	4	0.85
Perceived unfairness	4	0.82
Customer loyalty	6	0.88
Price sensitivity	4	0.79

Table 3: Questionnaire reliability analysis

Test indicator	Intensity of big data-driven price discrimination	Perceived unfairness	Customer loyalty	Price sensitivity
KMO value	0.81	0.77	0.83	0.78
Bartlett's test	432.12	298.45	567.34	385.00
Approximate chi-square				
Degrees of freedom	15	10	21	6
Significance (p-value)	0.000	0.000	0.000	0.000

2.2 Principles of the Deep Reinforcement Learning Algorithm

This study uses the Deep Q-Network (DQN) algorithm to train the dynamic pricing agent^[4]. DQN, a combination of deep learning and Q-Learning, uses a deep neural network to approximate the optimal action-value function $Q^*(s, a)$. Its core principles include three key components:

Q-Network: At its core is a deep neural network. The network takes the environment's state s as input and outputs Q-value estimates for all possible actions a , i.e., $Q(s, a; \theta)$, where θ are the network's weights. The agent formulates its policy by selecting the action with the highest Q-value.

Experience Replay: To break the correlation between data samples and improve data utilization, DQN introduces an experience replay mechanism. The experience tuples (s_i, a_i, r_i, s_{i+1}) generated by the agent's interaction with the environment are stored in a fixed-size "replay buffer." During training, the algorithm randomly samples a mini-batch of experiences from the buffer to update the network weights, making the training process more stable and efficient.

Target Network: To address the instability that can arise in Q-Learning from using the same network for both Q-value estimation and target Q-value calculation, DQN employs a dual-network structure. In addition to the online Q-network, there is a target network with the same architecture but with parameters that are

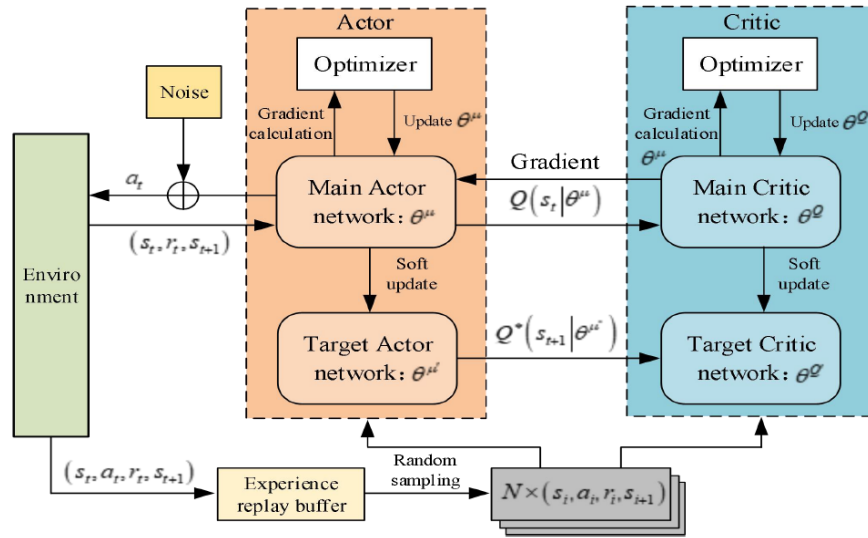


Figure 2: Deep Q-Network (DQN) Algorithm Structure Diagram.

Table 4: The key parameters used in the experiment.

Parameter	Description	Value
Episodes	The number of full interactions between the agent and the environment	500
Discount Factor (γ)	The weight of future rewards; a value close to 1 indicates a greater focus on long-term returns	0.99
Learning Rate	The step size for updating the neural network weights	0.001
Exploration Rate (ϵ)	The probability of the agent exploring new actions, linearly decaying from 1.0 to 0.05	1.0 -> 0.05
Replay Buffer Size	The maximum number of experience tuples to store	100,000
Batch Size	The number of samples drawn from the replay buffer for each training step	64
Target Network Update Freq.	The frequency (in steps) at which the main network weights are copied to the target network	100
Reward Function Params (α, β)	The penalty weights for loyalty decrease and customer churn	1.5, 5.0

updated more slowly. The target network is used to calculate the target Q-values. Its parameters are periodically copied from the main Q-network’s parameters, which helps to fix the update target and reduce oscillations during training.

The agent selects an action 'a' based on the current state 's' via the Q-network. After interacting with the environment, it receives a reward 'r' and a new state 's'. This experience is stored in the replay buffer. During training, samples are drawn from the replay buffer, and the target network is used to calculate the target Q-value to update the Q-network (Figure 2).

2.3 Experimental Procedure and Parameter Settings

The reinforcement learning experimental procedure in this study is rigorous, with specific operational details as follows:

Experimental Procedure:

Environment Construction: Based on the results of the questionnaire analysis, an e-commerce market environment with 10,000 simulated customers was constructed, with each customer having loyalty and price sensitivity attributes.

Agent Definition: The state space, action space, and reward function for the DQN agent were defined.

Training Loop: The agent was trained for 500 episodes in the simulated environment, interacting with all customers once per episode, learning and optimizing its pricing policy through the experience replay mechanism.

Policy Evaluation: The trained RL policy was compared with a greedy policy and a fixed-price policy to evaluate its performance in terms of total revenue and customer retention rate.

Parameter Settings: The key parameters used in the experiment are shown in the Table 4

Evaluation Metrics: To assess the effectiveness of different pricing policies, we used the following two core metrics:

Total Revenue: The total profit obtained by the platform over the entire test period.

Customer Retention Rate: The percentage of non-churned customers out of the total number of customers at the end of the test period.

2.4 Experimental Environment

All simulations were carried out on a workstation equipped with an Intel®Core™ i7-10700K @ 3.80 GHz CPU, NVIDIA®GeForce RTX™ 3070 (8 GB) GPU and 32 GB DDR4 RAM, running Ubuntu 20.04.3 LTS. The code was implemented in Python 3.8.10 and relied primarily on PyTorch 1.10.1, NumPy 1.21.5, Pandas 1.3.5, and Matplotlib 3.5.1.

3 Results and Analysis

3.1 Verification of Theoretical Model

Hypotheses

The regression analysis results strongly supported our initial hypotheses:

1.The intensity of big data-driven price discrimination has a significant negative impact on customer loyalty ($\beta = -0.28, p < 0.01$), verifying H1.

2.After introducing “perceived unfairness” as a mediating variable, the direct effect coefficient of price discrimination decreased significantly. The Sobel test Z-value was 2.13 ($p < 0.05$), and the mediating effect explained 42% of the total effect, indicating that perceived unfairness plays a significant mediating role, verifying H2.

3.The interaction term between price sensitivity and the intensity of price discrimination has a significant negative impact on customer loyalty ($\beta = -0.18, p < 0.05$), indicating that the higher the price sensitivity, the stronger the negative impact of price discrimination on loyalty, verifying H3.

3.2 Results of Reinforcement Learning Experiment

The DQN agent was trained and evaluated in the constructed simulation environment.

Learning Process Analysis: As shown in the training reward curve in Figure 3, the cumulative reward obtained by the agent in each episode steadily increased and began to converge after about 400 episodes. This indicates that the agent successfully learned an effective pricing policy through continuous interaction with the environment.

Policy Comparison Analysis: After training, we compared the RL policy with the two baseline policies (see Figure 4). The results show:

Total Revenue: RL Policy > Fixed-Price Policy > Greedy Policy.

Customer Retention Rate: RL Policy (approx. 95%) > Fixed-Price Policy (approx. 65%) > Greedy Policy (approx. 35%).

Although the greedy policy attempts to extract maximum profit in single transactions, its long-term total revenue is the lowest due to significant customer churn. The RL policy, by intelligently balancing revenue and customer relationships, achieved a win-win in both total revenue and customer retention.

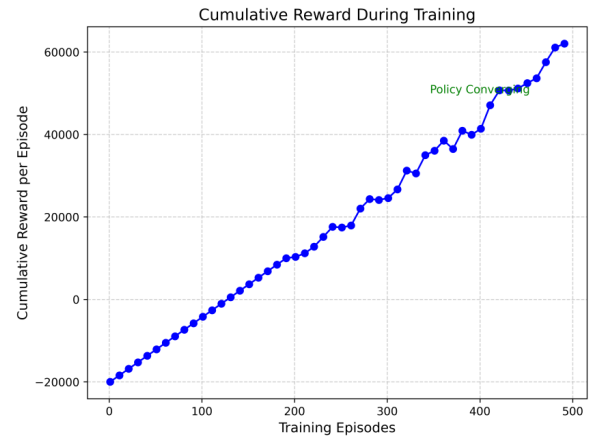


Figure 3: Reinforcement Learning Training Reward Curve

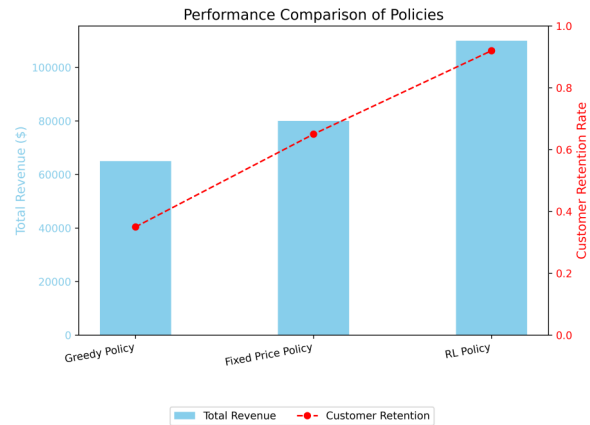


Figure 4: Performance Comparison of Different Pricing Policies

Final Learned Policy Analysis: Figure 5 reveals the details of the final pricing policy learned by the agent. The agent adopted significantly different actions for different types of customers:

For low-loyalty customers (yellow bars): The agent tended to offer discounts (-10% and -20%) to win back customers, increase their loyalty, and prevent them from churning.

For high-loyalty customers (purple bars): The agent mainly used the base price (0%) or a slight markup (+10%) to obtain steady profits without damaging customer trust. It almost never applied extreme markups (+20%) to loyal customers, as the reward function taught it that doing so would severely damage loyalty and be counterproductive.

4 Conclusions

This work employed a hybrid approach of empirical analysis and machine learning simulation to investigate the impact of dynamic pricing on customer loyalty in a big data environment. We first validated a theoretical model through questionnaire surveys and regression analysis. Subsequently, based on these findings, we

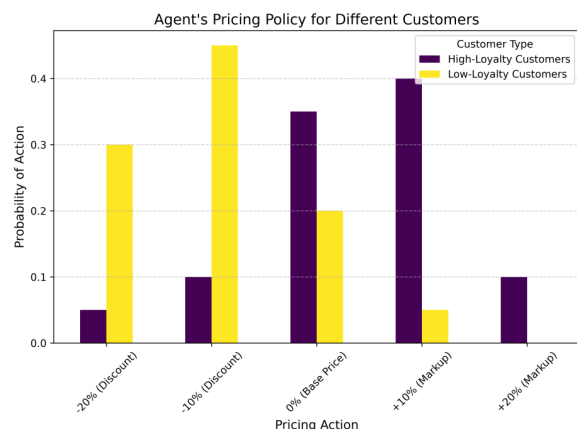


Figure 5: The Agent's Learned Pricing Policy for Different Customers

constructed a simulated market and utilized the Deep Q-Network (DQN) algorithm^[4] from deep reinforcement learning to train a pricing agent oriented towards long-term value.

The main findings are as follows: First, the empirical analysis reconfirmed that big data-driven price discrimination significantly harms customer loyalty by exacerbating users' "perceived unfairness,"^[6-10] an effect that is more pronounced among price-sensitive users^[8]. Second, the reinforcement learning simulation demonstrated that an intelligent pricing strategy oriented towards long-term customer value far outperforms myopic, greedy strategies in both total revenue and customer retention. Third, the optimal policy learned by the agent is not to "gouge loyal customers" but to "cherish customers"—that is, to apply moderate pricing to high-loyalty customers while proactively offering discounts to low-loyalty customers at risk of churning, thereby achieving a dynamic balance between business interests and customer relationships.

Looking forward, future research can be deepened in several aspects: First, by enhancing the external validity of the research through validation with larger, more diverse, and cross-cultural

samples. Second, by building more complex simulation environments that incorporate elements like competitive game dynamics. Third, by exploring more advanced algorithms, such as DDPG or Actor-Critic, to handle continuous action spaces, and by introducing more ethical evaluation metrics, such as a "Price Disparity Ratio" and a "Fairness Index," to ensure algorithmic social responsibility. Finally, conducting qualitative case studies and real-world A/B testing in collaboration with companies will be a key step in bridging the gap between theoretical models and business practice.

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